EXP NO: 1

Estimation of Characteristic Impedance (Zo), and Propagation Constant (γ) of a Transmission Line from Primary Constants.

Test Case 1:

Investigating the significance of Characteristic Impedance (Zo), and Propagation Constant

(γ) of a Transmission Line using the following parameters

Code:

```
clear; clc;

R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);

w=2*%pi*f;

Z=R+(%i*w*L);

Y=G+(%i*w*C);

Zo=sqrt(Z/Y);

C=round(real(Zo));

D=round(imag(Zo));

printf('-Zo = %f + j(%f)ohms\n',C,D);

P=sqrt(Z*Y);

a=real(P);

a1=round(a*10000)/10000;

printf('-Attenuation constant a = %f neper/km\n',a1);

b=imag(P);

b1=round(b*10000)/10000;

printf('-Phase constant b = %f radians/km',b1);
```

OUTPUT:

Simulation results			Theoretical results	
Z0(ohms)	α(neper/km)	β(radians/km)	α(neper/km)	β(radians/km)
	0.33	2.99	0.2378	4.421

THEORETICAL RESULTS:

Given,

$R = 10\Omega$

$C = 8\mu F$

$I = 3.5mH$

$C = 4\mu mho$

$F = 2khz$

$\omega = 2\pi f$

$= 2\pi \times 2\times10^{3}$

$\omega = 12566.37$

$$Z_o = \sqrt{\frac{R+j\omega L}{G+j\omega C}} = \sqrt{\frac{10+j(12566.37)(0.0035)}{4\times10^{6}+j(12566.37\times8\times10^{-6})}}$$

$$Z_o = \sqrt{\frac{10+j43.98}{0.000004+j0.10053}}$$

$$Z_o = 20.95 - j2.17\Omega$$

$$|Z_o| = \sqrt{(20.95)^2+(2.17)^2}$$

$$Z_o = 21.06\Omega$$

$$\gamma = \sqrt{(R+j\omega L)(G+j\omega C)}$$

$$\gamma = \sqrt{(10+j43.98)(0.000004+j0.10053)}$$

$$\gamma = 0.2378 + j4.421$$

$\alpha = 0.2378$ neper/km, $\beta = 4.421$ rad/km.

SIMULATION RESULTS:

Scilab code:

```
clear; clc;
R=10;L=0.0035;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*%pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```

Result:

```
-Zo = 15.000000 + j(-2.000000)ohms
-Attenuation constant a = 0.336000 neper/km
-Phase constant b = 2.992700 radians/km
```

TEST CASE 2:

Examine the impact of resistance and inductance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line at a frequency f=2.5 Khz. Vary the resistance value from 10 Ω to 50 Ω at the interval of 10 Ω with L,G and Cvalues constant and calculate the Characteristic Impedance and Propagation Constant .Vary the inductance values from 3.5 mH to 11.5 mH at the interval of 2.5 mH with R,G and C values constant.

PROGRAM:

```
clear; clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*%pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
```

OUTPUT:

Variable R

Constants L=3mH; f=2000; G=.4*(10^-6); C=8*(10^-6).

S.No	(R) (Ω)	Theoretical (Z_0) (Ω)	Theoretical α	Theoretical β	Simulation α	Simulation β
1	10	19.53 − j2.546	0.2560	1.9635	0.2560	1.9635
2	20	19.994 − j4.975	0.5002	2.0100	0.5002	2.0100
3	30	20.667 − j7.220	0.7258	2.0777	0.7258	2.0777
4	40	21.468 − j9.267	0.9316	2.1582	0.9316	2.1582
5	50	22.337 − j11.133	1.1192	2.2456	1.1192	2.2456

Variable L

Constants R=10Ω; f=2000; G=.4*(10^-6); C=8*(10^-6).

S.No	(L) (mH)	Theoretical (Z_0) (Ω)	Theoretical α	Theoretical β	Simulation α	Simulation β
1	3.5	21.050 − j2.363	0.2375	2.1161	0.2375	2.1161
2	8	31.662 − j1.571	0.1579	3.1830	0.1579	3.1830
3	6	27.446 − j1.812	0.1822	2.7592	0.1822	2.7592
4	11	37.105 − j1.340	0.1348	3.7302	0.1348	3.7302
5	11.5	37.937 − j1.311	0.1318	3.8138	0.1318	3.8138

SIMULATION RESULT:

Scilab code:

```
clear; clc;
R=20;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*%pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;

printf('-Phase constant b = %f radians/km',b1);
```

Result:

```
-Zo = 14.000000 + j(-4.000000)ohms
-Attenuation constant a = 0.707400 neper/km
-Phase constant b = 2.842600 radians/km
```

THEORETICAL RESULT:

Given,

$L = 3mH$

$f = 2000Hz$

$G = 0.4\times10^{-6}S$

$C = 8\times10^{-6}F$

$R = 10, 20, 30, 40, 50\Omega.$

(i) $\omega = 2\pi f = 2\times3.14\times2000 = 12566.37$ rad/s

$Z_0 = \sqrt{\dfrac{10+j(12566.37\times0.003)}{4\times10^{-6}+j(12566.37\times8\times10^{-6})}} = \sqrt{\dfrac{10+j37.69}{0.4\times10^{-6}+j0.100}}$

$Z_0 = 19.57$

$\gamma = \sqrt{(10+j37.69)(0.4\times10^{-6}+j0.10053)}$

$\alpha = 0.265$

$\beta = 3.78$

$\gamma = 0.265+j3.78$

(ii) $R = 20.$

$Z_0 = \sqrt{\dfrac{20+j37.69}{0.4\times10^{-6}+j0.10053}} \Rightarrow Z_0 = 20.34$

$\gamma = 0.510+j3.816$

$\alpha = 0.510$

$\beta = 3.186$

iii) $R = 30$

$Z_0 = \sqrt{\dfrac{30+j37.69}{0.4\times10^{-6}+j0.10053}}$

$Z_0 = 21.52\Omega$

$\gamma = 0.742+j3.865 \Rightarrow \alpha = 0.742$

$\beta = 3.862$

iv) $R = 40$

$Z_0 = 22.99\Omega$

$Z_0 = \sqrt{\dfrac{40+j37.69}{0.4\times10^{-6}+j0.10053}}$

$\gamma = 0.959+j3.930 \Rightarrow \alpha = 0.959$

$\beta = 3.930.$

v) $R = 50$

$Z_0 = \sqrt{\dfrac{50+j37.69}{0.4\times10^{-6}+j0.10053}}$

$\gamma = 1.162+j4.009$

$Z_0 = 24.64\Omega$

$\alpha = 1.162 \quad \beta = 4.009$

Given,

$R = 10\Omega, \quad f = 2000Hz$

$G = 0.4\times10^{-6} \quad C = 8\times10^{-6}$

i) $L = 3.5mH.$

$Z_0 = \sqrt{\dfrac{10+j43.98}{0.4\times10^{-6}+j0.10053}} = 21.07\Omega$

$\gamma = \sqrt{(10+j43.98)(0.4\times10^{-6}+j0.10053)} = \gamma = 0.248+j4.418$

$\alpha = 0.248 \quad \beta = 4.418$

ii) $L = 8mH$

$Z_0 = \sqrt{\dfrac{10+j106.81}{0.4\times10^{-6}+j0.10053}} = 32.86\Omega$

$\gamma = \sqrt{(10+j106.81)(0.4\times10^{-6}+j0.10053)} = 0.240+j7.151$

$\alpha = 0.240, \quad \beta = 7.151$

iii) $L = \cancel{8.5mH}\ 6mH$

$Z_0 = \sqrt{\dfrac{10+j75.40}{0.4\times10^{-6}+j0.10053}} = \cancel{32.86}\ 27.60\Omega$

$\gamma = \sqrt{(10+j75.40)(0.4\times10^{-6}+j0.10053)} = \gamma = 0.243+j6.010$

$\alpha = 0.243, \quad \beta = 6.010.$

iv) $L = 11mH.$

$Z_0 = \sqrt{\dfrac{10+j138.23}{0.4\times10^{-6}+j0.10053}} = 37.39\Omega$

$\gamma = \sqrt{(10+j138.23)(0.4\times10^{-6}+j0.10053)} = 0.238+j8.247$

$\alpha = 0.238, \quad \beta = 8.248$

v) $L = 11.5mH$

$Z_0 = \sqrt{\dfrac{10+j144.51}{0.4\times10^{-6}+j0.10053}} = 38.23\Omega$

$\gamma = \sqrt{(10+j144.51)(0.4\times10^{-6}+j0.10053)} = 0.238+j8.293.$

Test Case 3:

Explore the impact of frequency and capacitance on the significance of Characteristic Impedance (Zo) and Propagation Constant (γ) of a Transmission Line. Vary the frequency from 2KHz to 10 KHz at the interval of 2 KHz with L,R ,G and C values constant and calculate the Characteristic Impedance and Propagation Constant . Vary the capacitance values from 8 µF to 16 µF at the interval of 2 µF with R,G and L values constant.

OUTPUT:

Variable f

Constants L=3mH; f=2000; G=.4*(10^-6); C=8*(10^-6).

S.No	**Input (f) (KHz)**	**Theoretical (Z_0) (Ω)**	**Theoretical α (neper/km)**	**Theoretical β (rad/km)**	**Simulation α (neper/km)**	**Simulation β (rad/km)**
1	2	**19.532 − j2.546**	**0.2560**	**1.9635**	**0.2560**	**1.9635**
2	4	**19.407 − j1.281**	**0.2576**	**3.9021**	**0.2576**	**3.9021**
3	6	**19.384 − j0.855**	**0.2580**	**5.8460**	**0.2580**	**5.8460**
4	8	**19.376 − j0.642**	**0.2581**	**7.7914**	**0.2581**	**7.7914**
5	10	**19.372 − j0.513**	**0.2581**	**9.7373**	**0.2581**	**9.7373**

Variable c

Constants R=10Ω; f=2000; G=.4*(10^-6); C=8*(10^-6).

S.NO	**INPUT (C) (µF)**	**Theoretical (Z_0) (Ω)**	**Theoretical α**	**Theoretical β**	**Simulation α**	**Simulation β**
1	8	19.532 − j2.546	0.2560	1.9635	0.2560	1.9635
2	10	17.470 − j2.278	0.2862	2.1953	0.2862	2.1953
3	12	15.948 − j2.079	0.3135	2.4048	0.3135	2.4048

S.NO	INPUT (C) (μF)	Theoretical (Z_0) (Ω)	Theoretical α	Theoretical β	Simulation α	Simulation β
4	14	14.765 − j1.925	0.3387	2.5975	0.3387	2.5975
5	16	13.811 − j1.801	0.3620	2.7769	0.3620	2.7769

SIMULATION RESULT:

Scilab code:

```
clear; clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*%pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```

Result:

```
-Zo = 14.000000 + j(-2.000000)ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km
```

Test case 3

$f = 2KHz$

$$Z_0 = \sqrt{\frac{10 + j37.70}{0.4 \times 10^{-6} + j0.10053}}$$

$$Z_0 = 19.53 - j2.546\,\Omega$$

$$\gamma = \sqrt{(10 + j37.70)(0.4 \times 10^{-6} + j0.10053)} = \quad \alpha = 0.256 \quad \beta = 1.964$$

$f = 4KHz$

$\omega = 25132.74$

$$Z_0 = 19.407 - j1.281\,\Omega$$

$$\gamma = 0.258 + j3.902$$

$f = 6KHz$

$\omega = 37699.11$

$$Z_0 = \sqrt{\frac{10 + j113.10}{0.4 \times 10^{-6} + j0.30159}}$$

$$Z_0 = 19.384 - j0.855\,\Omega$$

$$\gamma = 0.258 + j\,5.846 \text{ rad/Km}$$

$f = 8KHz$

$\omega = 50265.48$

$$Z_0 = 19.376 - j0.642\,\Omega$$

$$\gamma = 0.258 + j\,7.791$$

$f = 10KHz$

$\omega = 62831.85$

$$Z_0 = 19.372 - j0.513\,\Omega$$

$$\gamma = 0.258 + j\,9.373 \text{ rad/Km}$$